SHIVAM KAMBOJ

Quantum Control, QCVV & Device Modeling Physicist

Merced, CA · (408) 858-3104 · shivam1kamboj@gmail.com · Portfolio · GitHub: @shivam1kamboj

Quantum Control, QCVV & Hardware Modeling Physicist (Ph.D. Expected Fall 2026): Bridges first-principles circuit-QED and cavity-magnon Hamiltonians with physical hardware control and calibration. Experienced in pulse-level superconducting qudit control and Linear-Inversion Quantum Process Tomography at **LLNL**, COMSOL RF/AC-DC parameter extraction, experimental quantum optics collaboration, and production Python software (`mypy`, `ruff`, CI/CD).

HARDWARE CONTROL, QCVV & SIMULATION STACK

From Device Spec to Calibration Pipeline

- **Quantum Control & QCVV:** Superconducting qudit pulse-level control, optimal control gate evaluation, Linear-Inversion Quantum Process Tomography (LI-QPT), Moore-Penrose pseudo-inverse process reconstruction, automated Ramsey & T_1/T_2 fitting, $1/f$ noise characterization.
- **Electromagnetic & Device Modeling:** COMSOL RF & AC/DC finite-element simulation, capacitance & impedance matrix extraction, eigenmode & inductive participation analysis, circuit-QED quantization, coupled YIG-cavity magnon-photon modeling.
- **Scientific Software & HPC:** Python, SciPy, NumPy, QuTiP, Qiskit, Stim, PyMatching, Matplotlib; SLURM HPC clusters, Linux, Git, CI/CD, `pytest`, `mypy --strict`, `ruff`, Flask, Google Gemini API, GCP.

QUANTUM HARDWARE, CONTROL & MODELING EXPERIENCE

National Lab & Experimental Collaboration

Lawrence Livermore National Laboratory (LLNL) — *Physics Graduate Intern, Quantum Coherent Device Group* **May – Aug 2024**

- **Pulse-Level Qudit Control:** Developed modular Python software for pulse-level manipulation of multilevel superconducting qudits ($|1\rangle \rightarrow |3\rangle$ transitions), evaluating optimal control gate sequences and benchmarking hardware gate fidelities.
- **Linear-Inversion Quantum Process Tomography (LI-QPT):** Engineered a generalized linear-inversion tomography pipeline utilizing pseudo-inverse reconstruction to translate noisy experimental readout directly into quantum process maps.
- **Automated Coherence & Calibration Loops:** Built automated Ramsey fringe and T_1/T_2 relaxation fitting pipelines to characterize device drift and deliver rapid quantitative feedback for closed-loop control experiments.

Tian Group, University of California, Merced — *Graduate Student Researcher (Advisor: Prof. Lin Tian)* **May 2023 – Present**

- **COMSOL RF & AC/DC Parameter Extraction:** Built finite-element electromagnetic models of a proposed superconducting quantum Rabi dimer realization; extracted capacitances, mode frequencies, impedances, and inductive participation ratios to map physical layout constraints onto circuit Hamiltonians.
- **Open-System Decoherence & HPC Simulation:** Simulated dissipative Lindblad state evolution, photon loss, and macroscopic entanglement in QuTiP and sparse Exact Diagonalization solvers across checkpointed SLURM cluster sweeps.
- **Bosonic Qubit Stabilization:** Combined normal-mode analysis and parity-resolved diagonalization to identify level crossings and stabilize parity-protected cat-state doublets under realistic device parameters.

Sharping Group (UC Merced) & Hurst Group (SJSU / UC Merced) — *Experimental Collaborator & Researcher* **Jun 2021 – Present**

- **Experimental Cavity-Magnon & Circuit Realization (Sharping Group):** Formulate first-principles Hamiltonians for coupled YIG-cavity systems under parametric driving and collaborate with experimentalists on superconducting circuit realization and microwave measurement design.
- **Driven Magnetic Oscillator Arrays (Hurst Group):** Derived analytical dispersion relations and numerical solvers for 2D spin-torque nano-oscillator arrays, proving edge-mode robustness to hardware disorder (first-author in *Phys. Rev. B*).

PRODUCTION SOFTWARE & BENCHMARKING PIPELINES

PyPI Package · Type-Checked Python

`nonmarkov_qec` — **Correlated Hardware Noise & Surface-Code Benchmarking Suite (Released on PyPI) 2026 – Present**

- Built an open-source benchmarking pipeline combining Ornstein-Uhlenbeck and $1/f$ temporally correlated dephasing models, Stim circuit sampling, PyMatching decoding, and bootstrap Monte Carlo analysis to quantify non-Markovian hardware noise penalties.
- Enforced industry-standard software hygiene with 74 automated tests (`pytest`), strict static typing (`mypy --strict`), linting (`ruff`), and GitHub Actions CI/CD release automation.

Physics Concept Analyzer Tool (PCAT) — Cloud AI Application (Python, Flask, Gemini API, GCP) 2025

- Architected and deployed a full-stack application on Google App Engine automating structured physical model verification.

EDUCATION, PUBLICATIONS & PRESENTATIONS

Graduating Fall 2026

Ph.D. in Physics, UC Merced (Expected Fall 2026, Advisor: Prof. Lin Tian) · **M.S. in Physics**, San Jose State University (2022)

Selected Outputs: (1) **S. Kamboj** et al., *Phys. Rev. B* **109**, 094436 (2024); (2) **S. Kamboj**, J. Beurkens, J. Sharping, and L. Tian, "Superconducting realization of the Quantum Rabi Dimer" (In prep, 2026); (3) **S. Kamboj**, Y. Cho, and Y. J. Rosen, "Improving Fidelity of Quantum Gates Using Optimal Control," *LLNL Poster Symposium* (2024).